Electrical Circuit (1)

Introduction (week1 class2)

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Ohm's Law

$$v \propto i$$

$$v = iR$$

$$R = \rho \frac{\ell}{A}$$

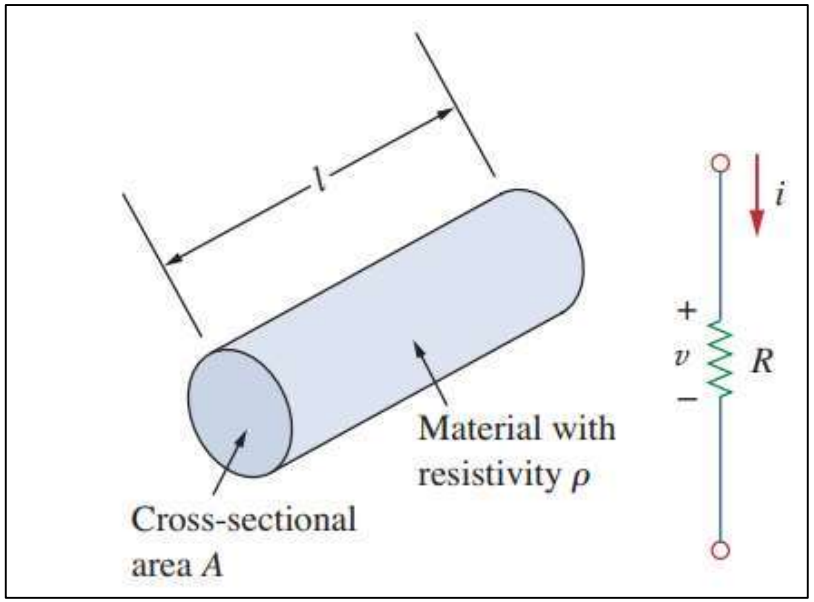
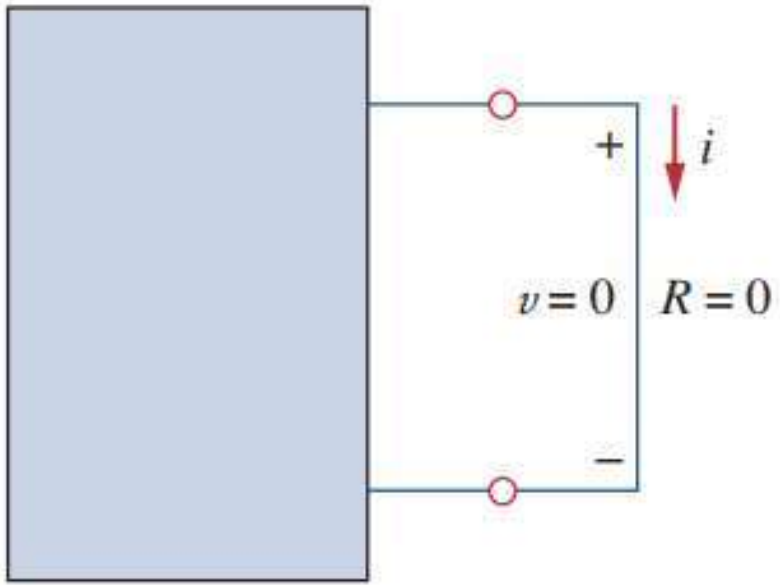


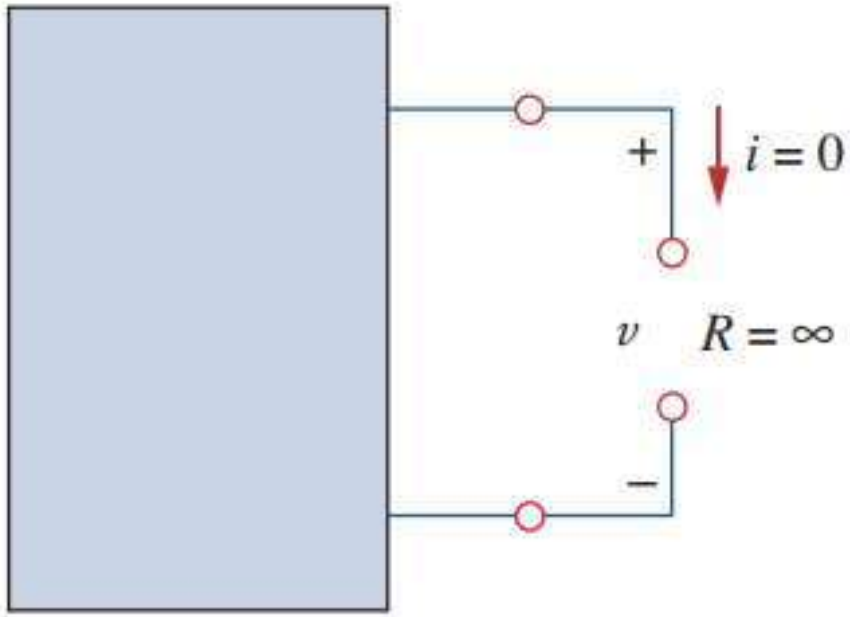
TABLE 2.1

Resistivities of common materials.

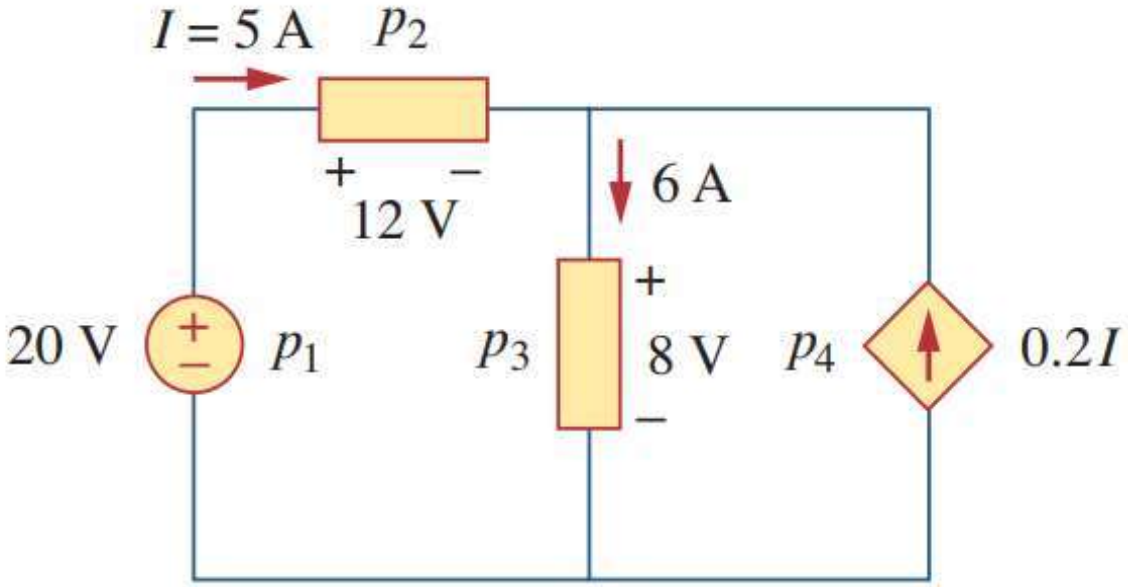
Material	Resistivity (Ω·m)	Usage
Silver	1.64×10^{-8}	Conductor
Copper	1.72×10^{-8}	Conductor
Aluminum	2.8×10^{-8}	Conductor
Gold	2.45×10^{-8}	Conductor
Carbon	4×10^{-5}	Semiconductor
Germanium	47×10^{-2}	Semiconductor
Silicon	6.4×10^{2}	Semiconductor
Paper	10^{10}	Insulator
Mica	5×10^{11}	Insulator
Glass	10^{12}	Insulator
Teflon	3×10^{12}	Insulator

Ohm's Law





Power and Energy



P1 = 20V*(-5A) = -100 watt

P2 = 12V*(5A) = 60 watt

P3 = 8V*(6A) = 48 watt

P4 = 8V*(-0.2*5A) = -8 watt

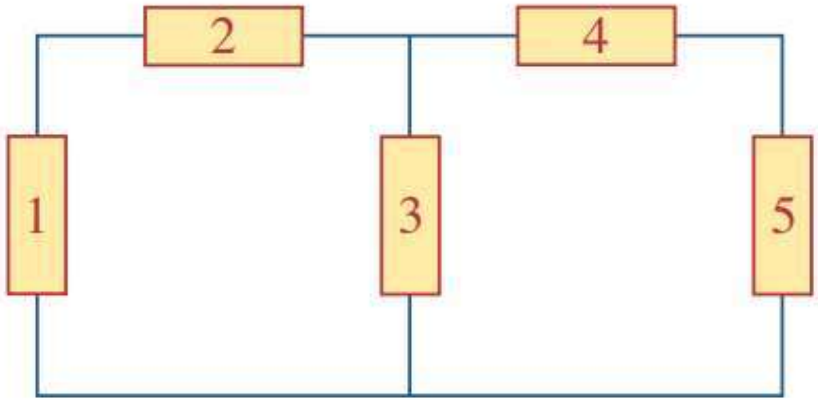
TABLE 1.3

Typical average monthly consumption of household appliances.

Appliance	kWh consumed	Appliance	kWh consumed
Water heater	500	Washing machine	120
Freezer	100	Stove	100
Lighting	100	Dryer	80
Dishwasher	35	Microwave oven	25
Electric iron	15	Personal computer	12
TV	10	Radio	8
Toaster	4	Clock	2

Power and Energy

1.17 Figure 1.28 shows a circuit with five elements. If $p_1 = -205$ W, $p_2 = 60$ W, $p_4 = 45$ W, $p_5 = 30$ W, calculate the power p_3 received or delivered by element 3.



†Nodes, Branches, and Loops

A **branch** represents a single element such as a voltage source or a resistor.

A **node** is the point of connection between two or more branches.

A **loop** is any closed path in a circuit.

$$b = l + n - 1$$

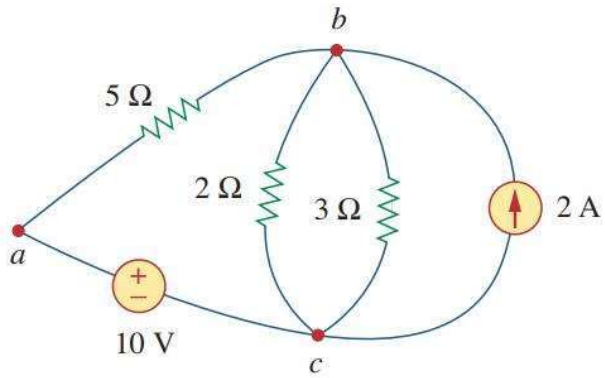
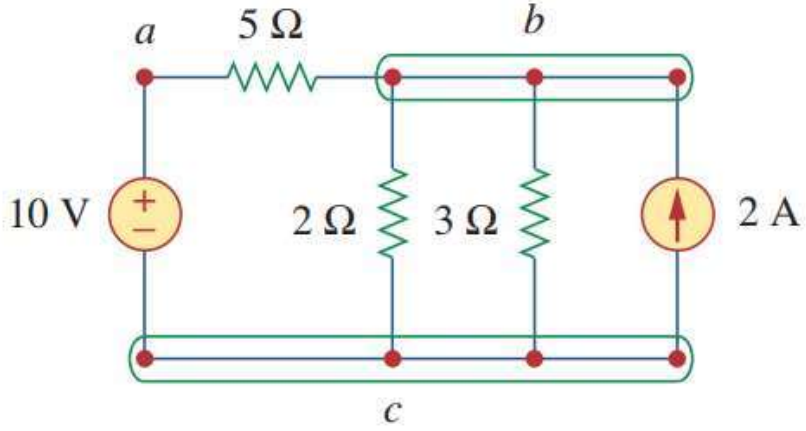


Figure 2.11
The three-node circuit of Fig. 2.10 is redrawn.



†Nodes, Branches, and Loops

Nodes

Branches

Loops

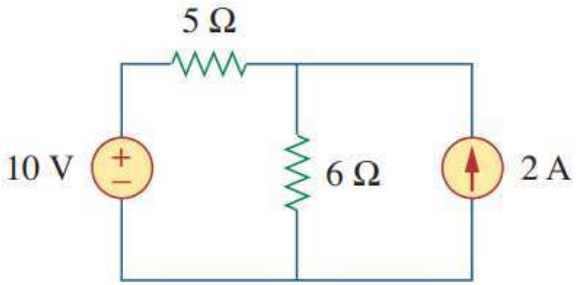


Figure 2.12
For Example 2.4.

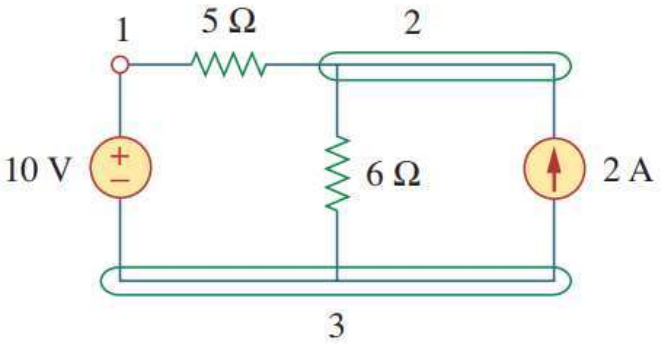


Figure 2.13
The three nodes in the circuit of Fig. 2.12.

†Nodes, Branches, and Loops

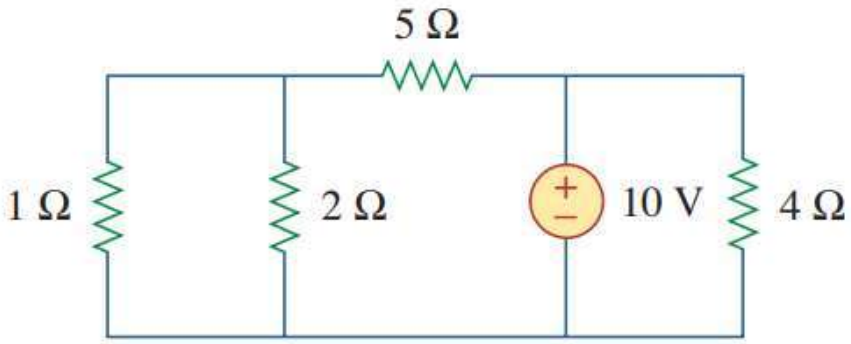


Figure 2.14
For Practice Prob. 2.4.

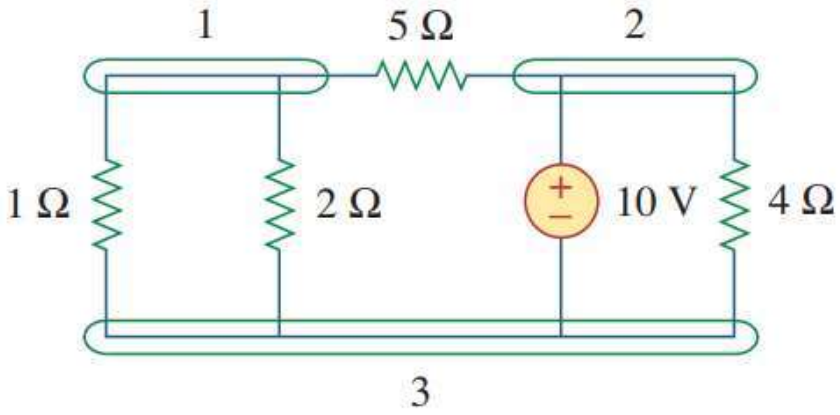


Figure 2.15
Answer for Practice Prob. 2.4.

Nodes

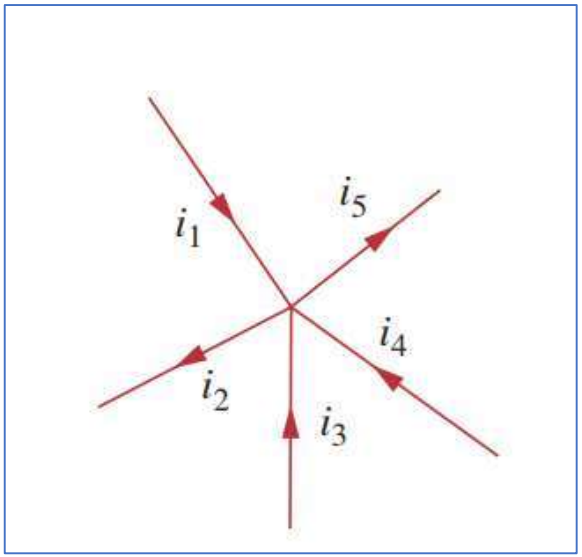
Branches

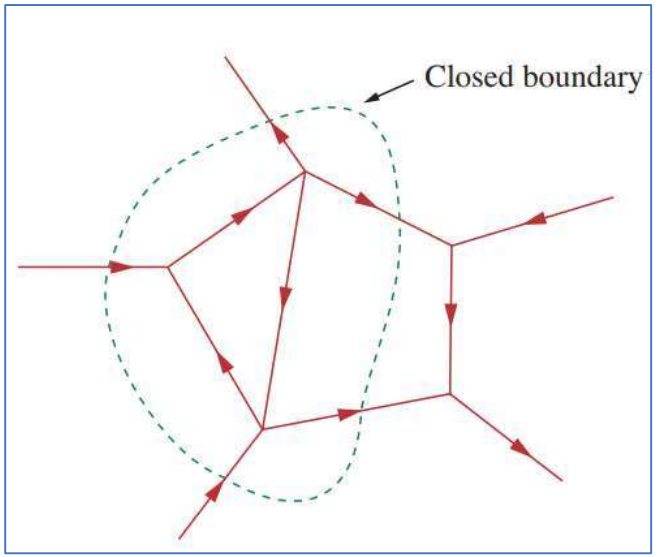
Loops

Kirchhoff's Laws

Kirchhoff's current law (KCL) states that the algebraic sum of currents entering a node (or a closed boundary) is zero.

$$\sum_{n=1}^{N} i_n = 0$$

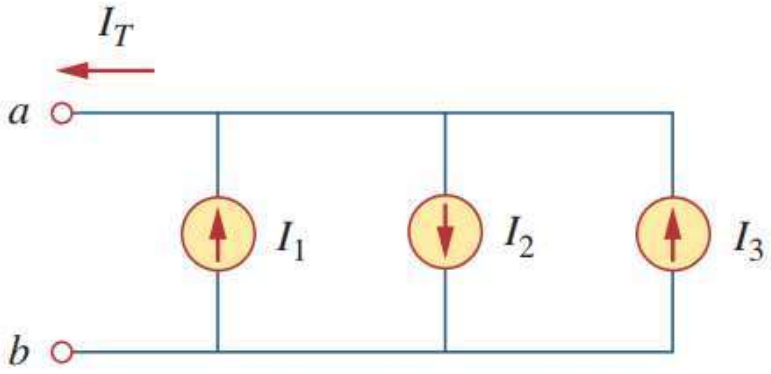


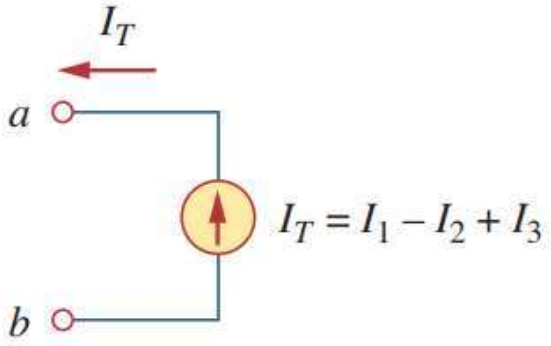


Kirchhoff's Laws

$$I_T + I_2 = I_1 + I_3$$

$$I_T = I_1 - I_2 + I_3$$





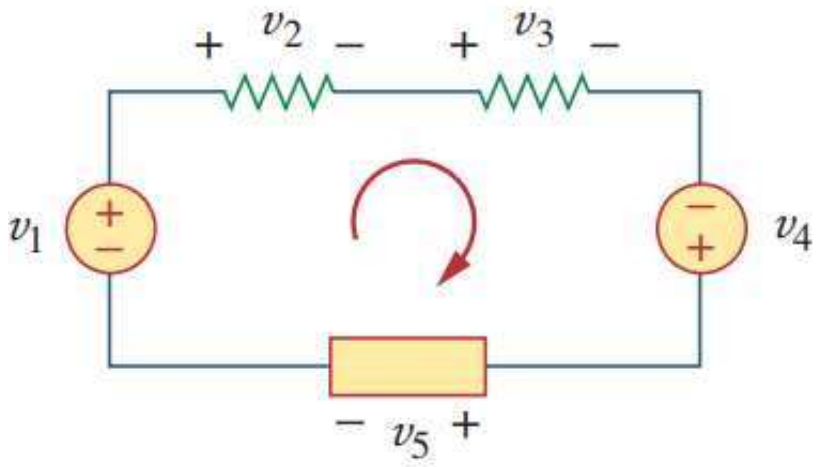
Kirchhoff's Laws

Kirchhoff's voltage law (KVL) states that the algebraic sum of all voltages around a closed path (or loop) is zero.

$$\sum_{m=1}^{M} v_m = 0$$

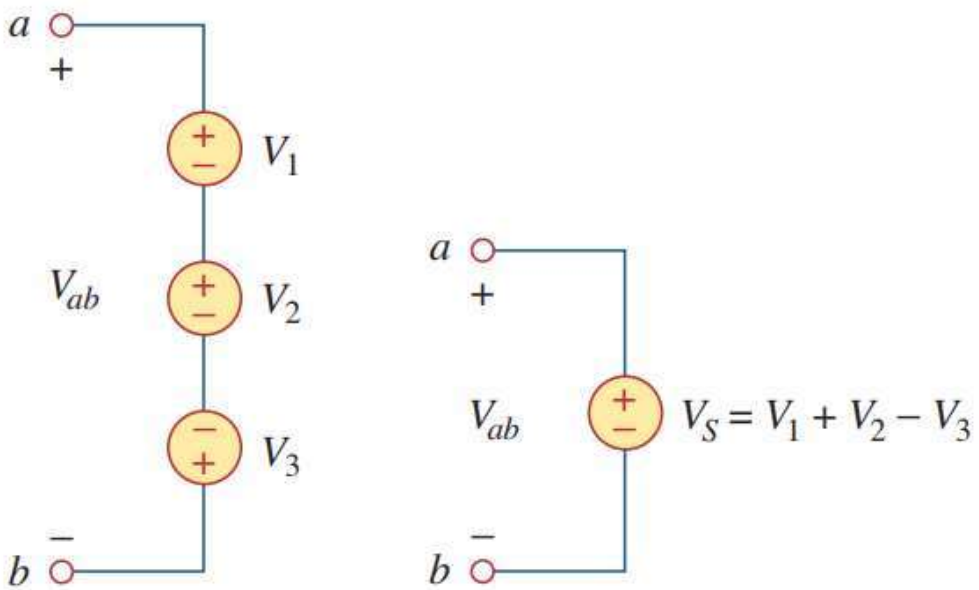
$$-v_1 + v_2 + v_3 - v_4 + v_5 = 0$$

$$v_2 + v_3 + v_5 = v_1 + v_4$$

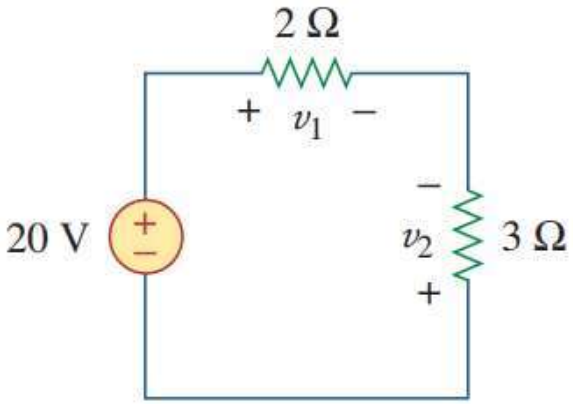


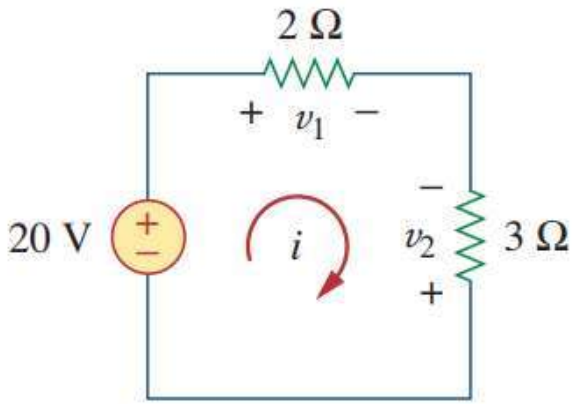
Kirchhoff's Laws

$$V_{ab} = V_1 + V_2 - V_3$$



Kirchhoff's Laws





From Ohm's law,

$$v_1 = 2i, \qquad v_2 = -3i \tag{2.5.1}$$

Applying KVL around the loop gives

$$-20 + v_1 - v_2 = 0 \tag{2.5.2}$$

Substituting Eq. (2.5.1) into Eq. (2.5.2), we obtain

$$-20 + 2i + 3i = 0 \quad \text{or} \quad 5i = 20 \quad \Rightarrow \quad i = 4\ \text{A}$$

Substituting i in Eq. (2.5.1) finally gives

$$v_1 = 8\ \text{V}, \qquad v_2 = -12\ \text{V}$$



Electrical Circuit (1)

Introduction (week2 class1)

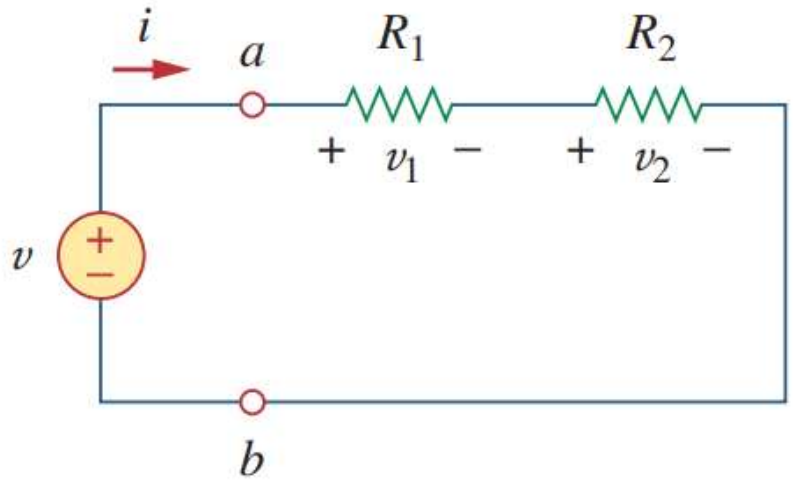
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Series Resistors and Voltage Division



$$v_1 = iR_1, \qquad v_2 = iR_2$$

$$-v + v_1 + v_2 = 0$$

$$v = v_1 + v_2 = i(R_1 + R_2)$$

$$i = \frac{v}{R_1 + R_2} \qquad R_{eq} = R_1 + R_2 \qquad v = iR_{eq}$$

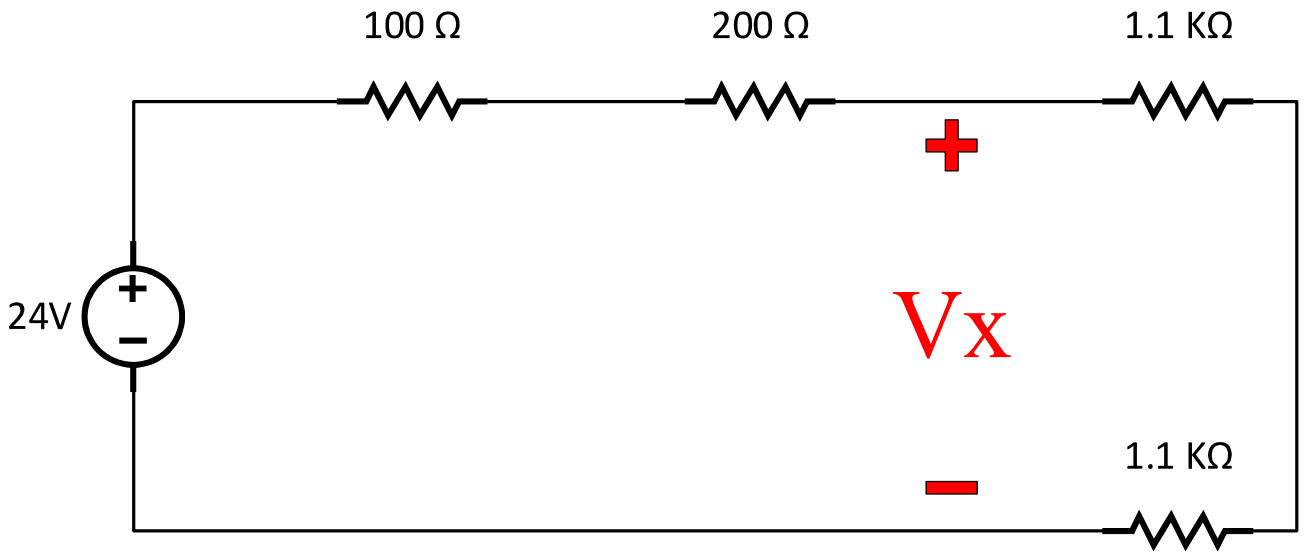
$$R_{eq} = R_1 + R_2 + \cdots + R_N = \sum_{n=1}^{N} R_n$$

$$v_1 = \frac{R_1}{R_1 + R_2} v, \qquad v_2 = \frac{R_2}{R_1 + R_2} v$$

$$v_n = \frac{R_n}{R_1 + R_2 + \cdots + R_N} v$$

Series Resistors and Voltage Division

Find Vx?

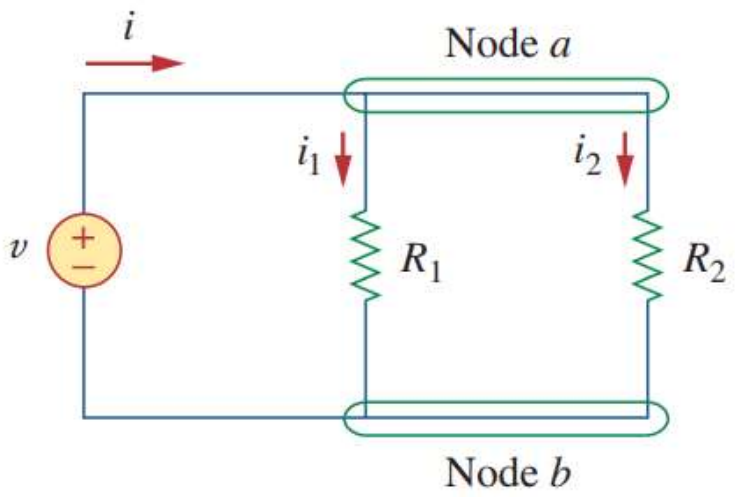


$$V_x = V_s \frac{R_3 + R_4}{R_1 + R_2 + R_3 + R_4}$$

$$V_x = 24 \frac{1100 + 1100}{100 + 200 + 1100 + 1100} = 17.6\ V$$

Parallel Resistors and Current Division

$$v = i_1 R_1 = i_2 R_2 \qquad i_1 = \frac{v}{R_1}, \qquad i_2 = \frac{v}{R_2}$$



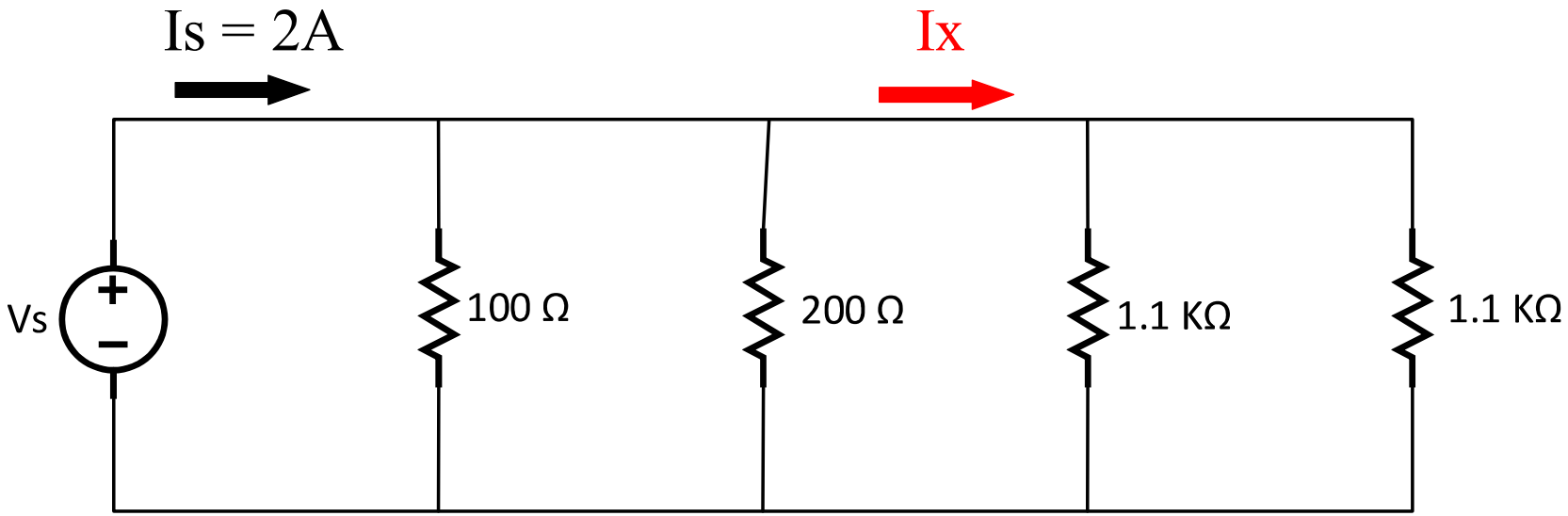
$$i = i_1 + i_2 \qquad i = \frac{v}{R_1} + \frac{v}{R_2} = v\left(\frac{1}{R_1} + \frac{1}{R_2}\right) = \frac{v}{R_{eq}}$$

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} \qquad \frac{1}{R_{eq}} = \frac{R_1 + R_2}{R_1 R_2} \qquad R_{eq} = \frac{R_1 R_2}{R_1 + R_2} \qquad \frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \cdots + \frac{1}{R_N}$$

$$v = iR_{eq} = \frac{iR_1 R_2}{R_1 + R_2} \qquad i_1 = \frac{R_2\, i}{R_1 + R_2}, \qquad i_2 = \frac{R_1\, i}{R_1 + R_2}$$

Parallel Resistors and Current Division

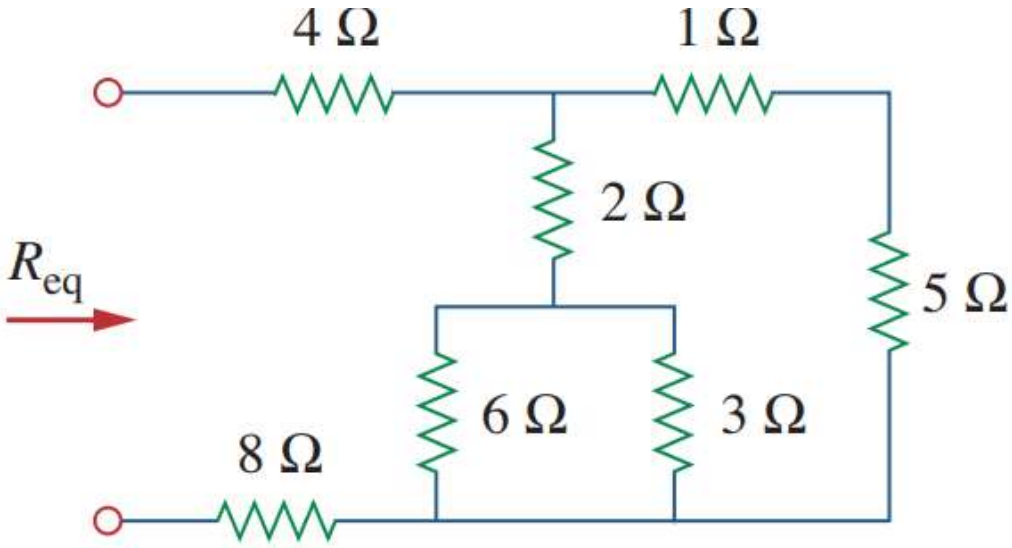
Find Ix?

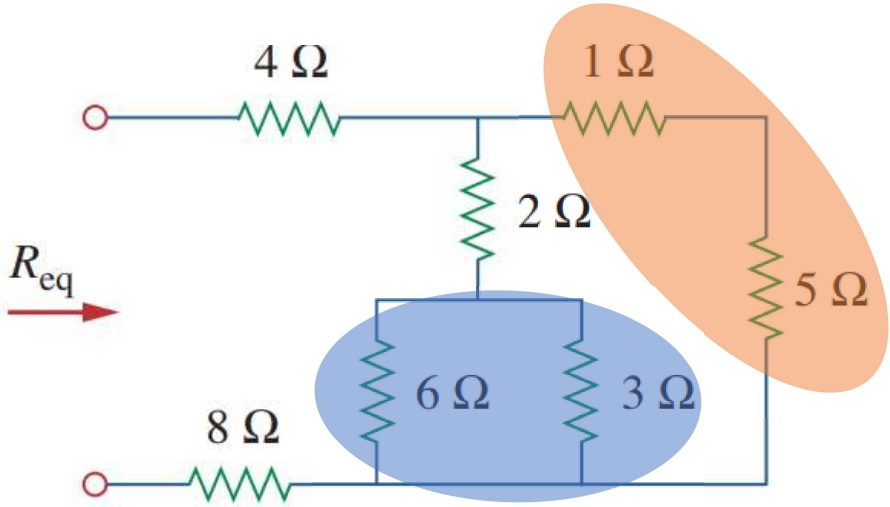


$$I_x = I_s \frac{R_{3||4}}{R_{1||2} + R_{3||4}}$$

$$I_x = 2 \frac{\frac{2000}{300}}{\frac{2000}{300} + 550} = 1.976\ A$$

Find R_{eq}?





Find R_{eq}?

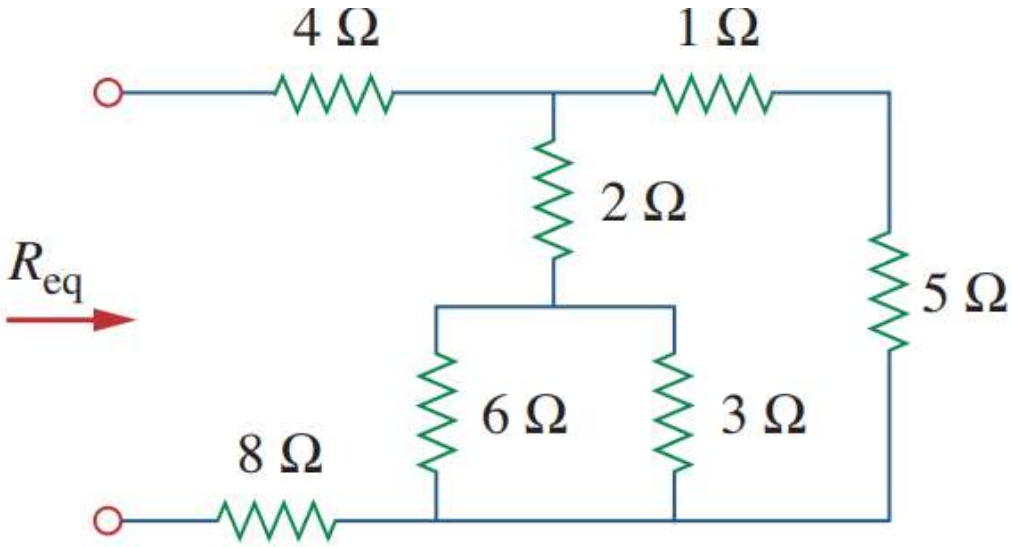
$$6\,\Omega \parallel 3\Omega = \frac{6 \times 3}{6 + 3} = 2\,\Omega$$

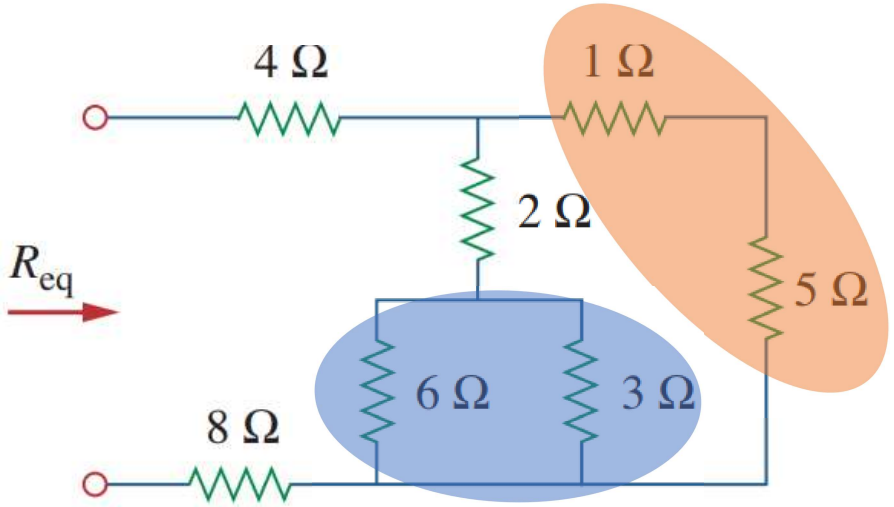
$$1\,\Omega + 5\,\Omega = 6\,\Omega$$

$$2\,\Omega + 2\,\Omega = 4\,\Omega$$

$$4\,\Omega \parallel 6\,\Omega = \frac{4 \times 6}{4 + 6} = 2.4\,\Omega$$

$$R_{eq} = 4\,\Omega + 2.4\,\Omega + 8\,\Omega = 14.4\,\Omega$$







Electrical Circuit (1)

Introduction (week2 class2)

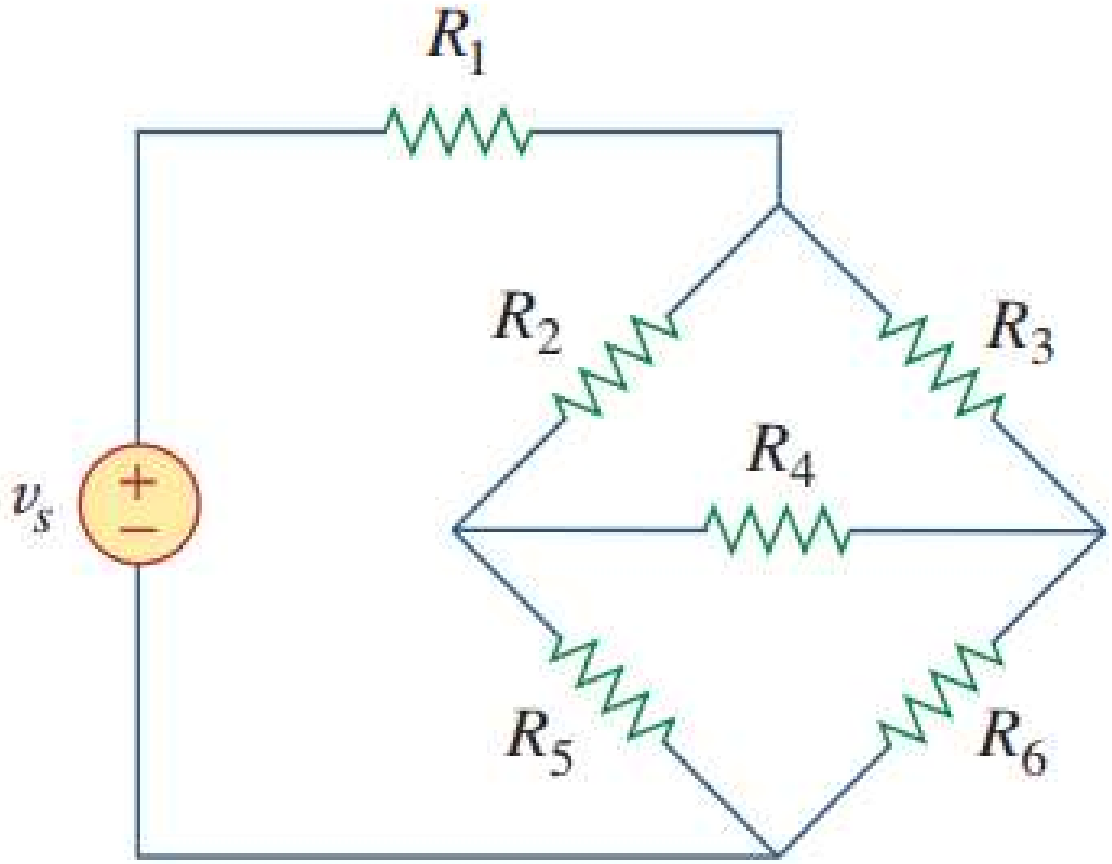
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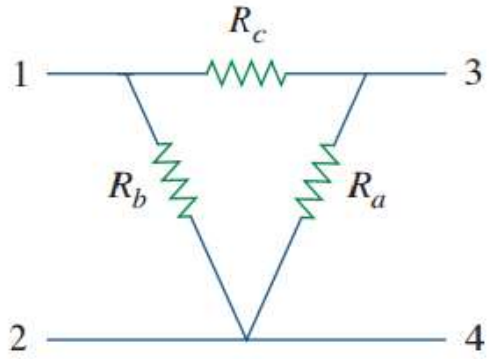
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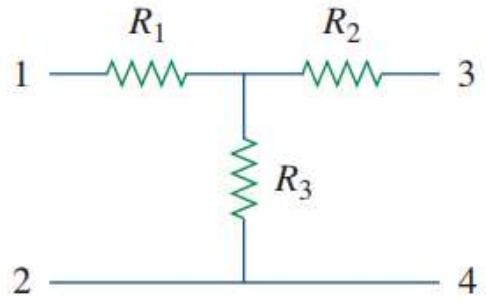
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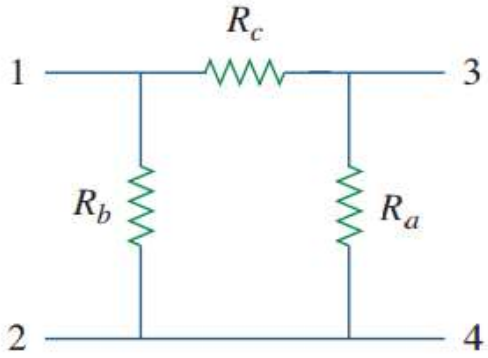
Wye-Delta Transformations

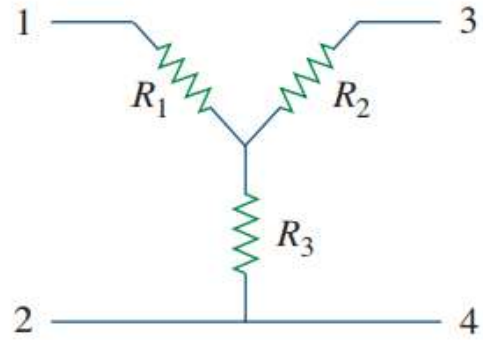


Wye-Delta Transformations







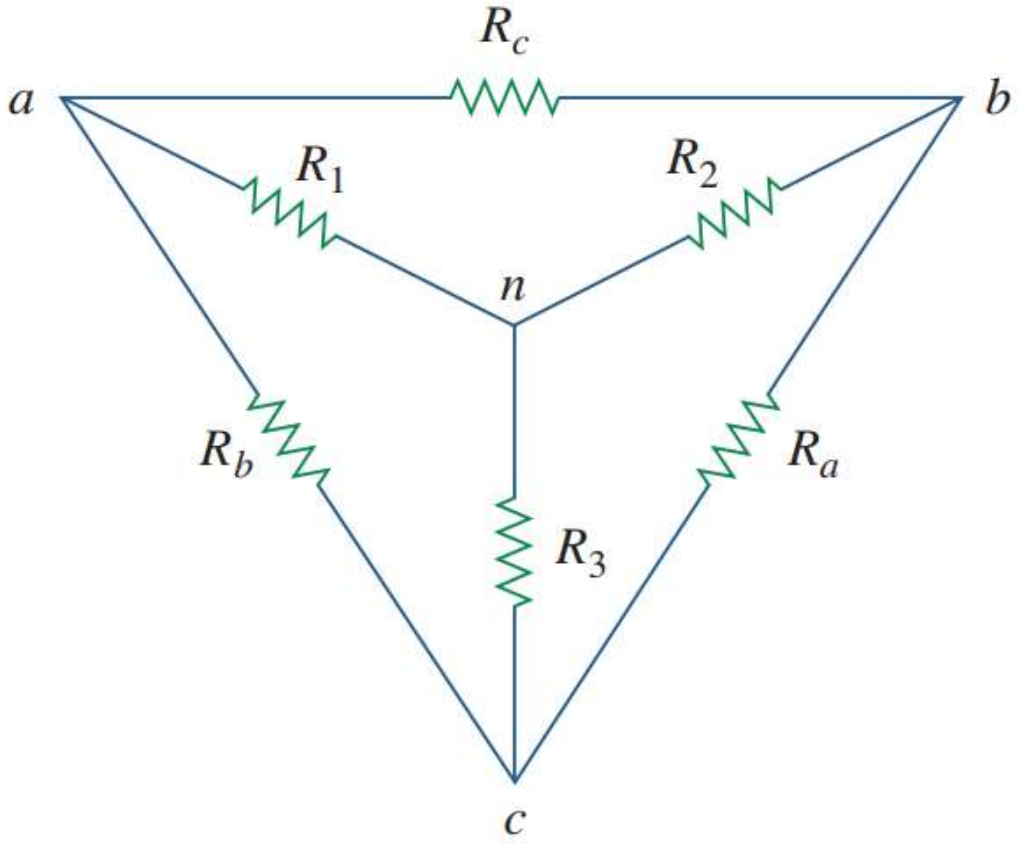


Wye-Delta Transformations

$$R_1 = \frac{R_b R_c}{R_a + R_b + R_c}$$

$$R_2 = \frac{R_c R_a}{R_a + R_b + R_c}$$

$$R_3 = \frac{R_a R_b}{R_a + R_b + R_c}$$

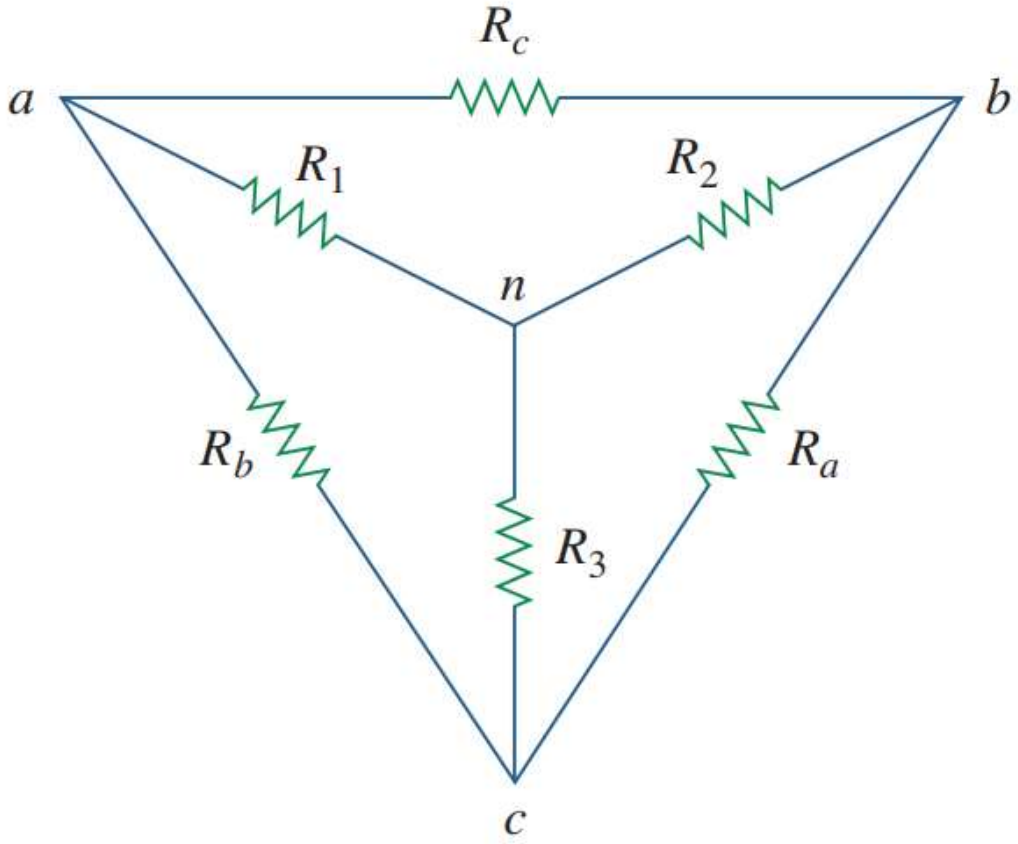


Wye-Delta Transformations

$$R_a = \frac{R_1R_2 + R_2R_3 + R_3R_1}{R_1}$$

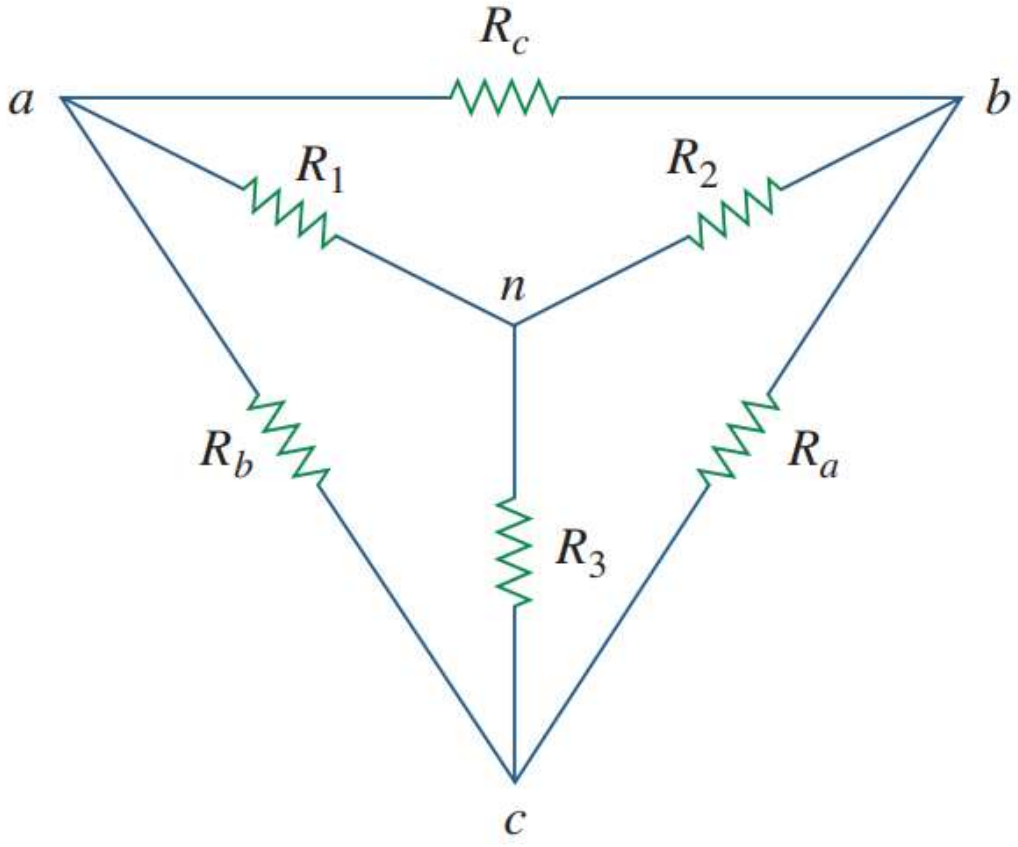
$$R_b = \frac{R_1R_2 + R_2R_3 + R_3R_1}{R_2}$$

$$R_c = \frac{R_1R_2 + R_2R_3 + R_3R_1}{R_3}$$

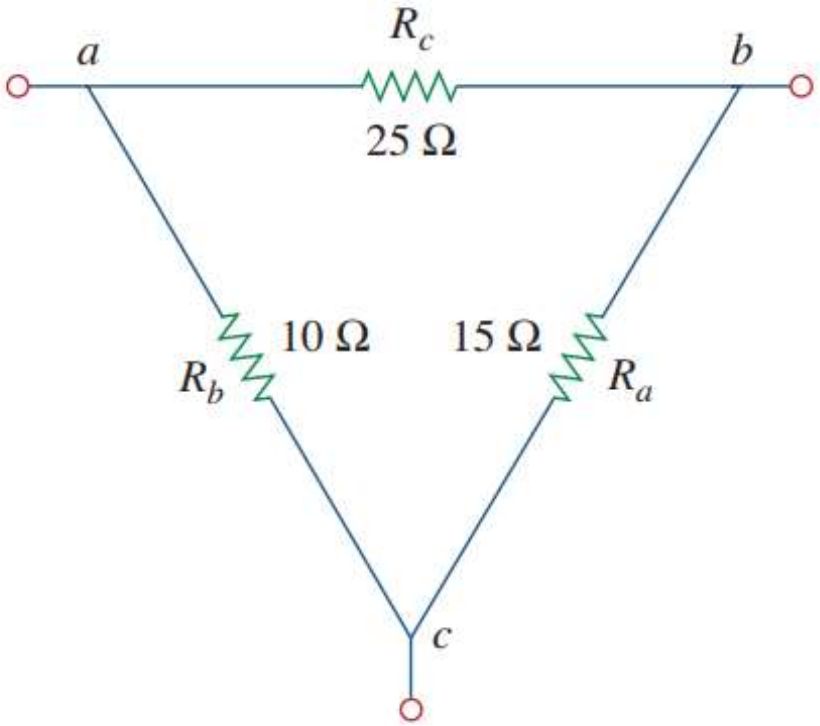


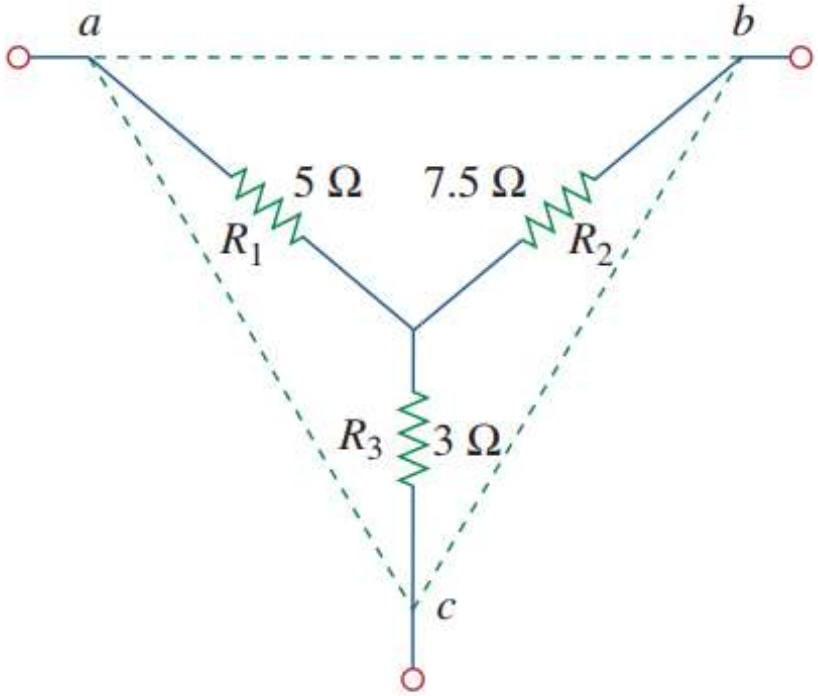
Wye-Delta Transformations

$$R_Y = \frac{R_\Delta}{3} \qquad \text{or} \qquad R_\Delta = 3R_Y$$



Wye-Delta Transformations

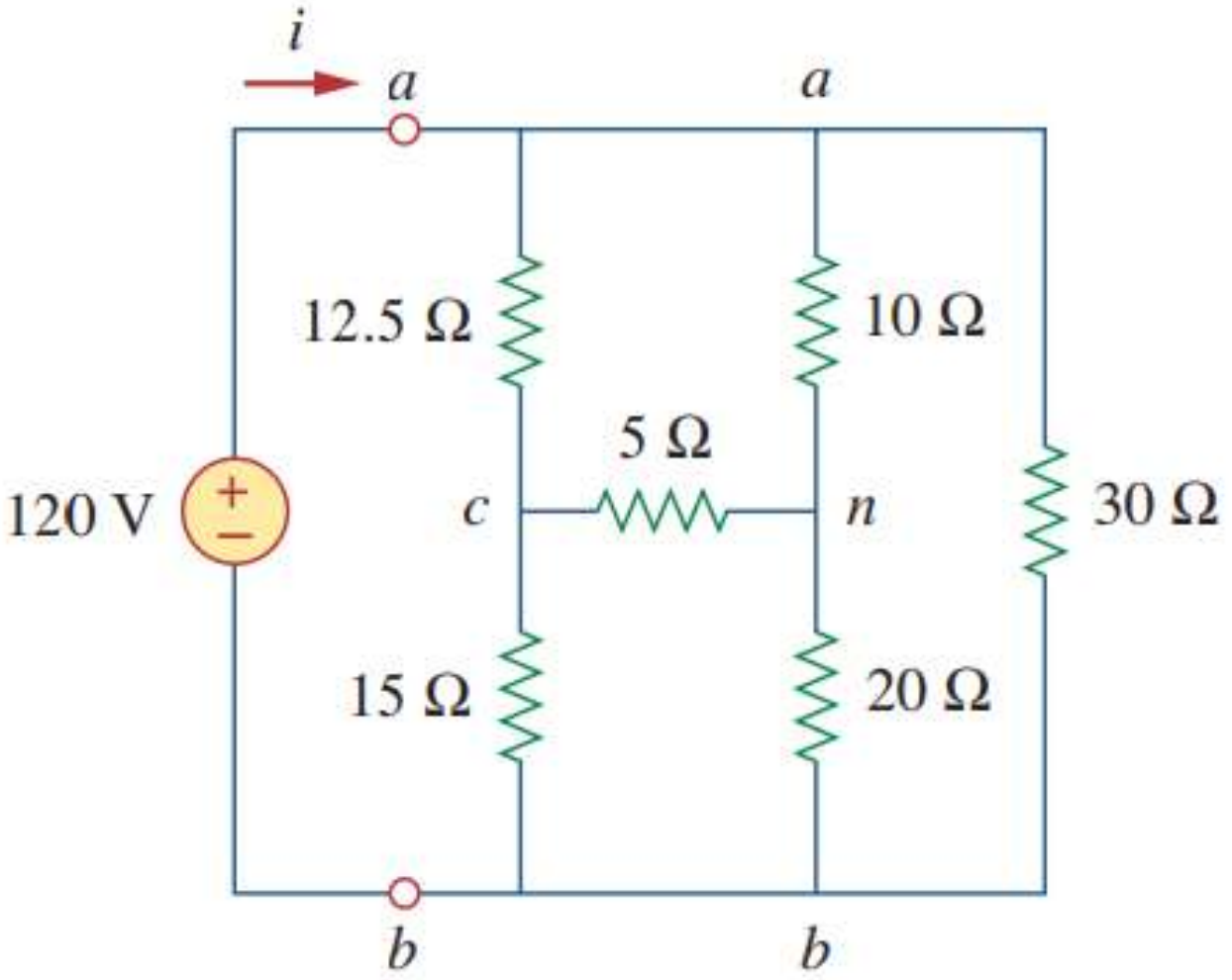




† Wye-Delta Transformations

Find R_{ab}

Find all deltas and stars



Quiz (Moodle) Find R_{ab}

